PRO1 – POSITIVE COMPACT PROBES AND FREE RESPONSE COEFFICIENTS

Abstract. This focused outline accompanies the eighteen-page free-response manuscript PRM1. It separates the continuum construction of a positive compact probe from the chamber fourth-trace comparison and the polarization cancellation needed for ordinary determinant coefficients. It retains the positive free mass, exact sampling and normalization conventions, error rates and parameter dependence.

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1. Introduction and target statement

PRO1 treats the free-response part of QEO1. The target combines PRM1 Theorems A and B. It assumes no interacting trajectory or mass tuning. All non-setup boxes have a proof drafted in PRM1 at the linked locators. The overall proof difficulty is 4/5; the target box rates only the final combination after its incoming results.

Overall plan: QEO1/T04. Proof manuscript: PRM1 Theorem B.

P. Positive compact probes and uniform free response

Model-specific. Overview Target.

Setting. Use the explicit chamber regulator and sampling in D. The four-by-four Euclidean Clifford matrices are $\gamma_\nu = \operatorname{diag}(\sigma_\nu, -\sigma_\nu)$, with Pauli matrices σ_ν. Fix $0 < \rho < 1$, $0 < \bar{m} \le 1$, $0 < \eta \le 1$ and $n = R/\eta \in \mathbb{N}$, $n \ge 2$. The lattice free mass is the strictly positive number $m = \bar{m}\eta$. Spinors are antiperiodic in all three directions. For real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, put $a = d^*\beta$ and $f = dd^*\beta$. Let D_β be the least side of a coordinate cube containing $\operatorname{supp}\beta$. For the midpoint-sampled, periodized bond probe a_η, set

$$B_t(c, c + e_\nu) = t\eta a_{\eta,\nu}(\eta c), \qquad N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}.$$

Here D^{ch} takes the absorbed phase, not an unabsorbed connection. Define ordinary derivatives and the logarithmic fourth coefficient separately:

$$c_j = N_0^{(j)}(0), \qquad b_{\mathrm{ch}} = (\log N_0)^{(4)}(0), \qquad \log N_0(0) = 0.$$

On $\mathbb{R}^3$, let $S_{\bar m} = (\bar m + \gamma\cdot\partial)^{-1}$. On $T_R = \mathbb{R}^3/R\mathbb{Z}^3$, let S^R be the same multiplier on antiperiodic momenta, and let $a^R = \sum_z a(\cdot + Rz)$. Put

$$b(f) = -6\,\mathrm{Tr}(S_{\bar m} i\gamma\cdot a)^4, \qquad b_R(f) = -6\,\mathrm{Tr}(S^R i\gamma\cdot a^R)^4.$$

Target theorem. For $R \ge D_\beta + 1$, the operators in these two fourth powers belong to the Schatten class of order four, and their fourth-power traces are real. For every fixed $\bar m > 0$, a real compact smooth two-form β can be chosen so that $b(dd^*\beta) > 0$. For each real nonzero compact smooth probe, regardless of this sign, there are constants with

$$|b_{\mathrm{ch}}(\eta, R) - b_R(f)| \le C_{\rho,\bar m,\beta}\eta^{1/4}, \qquad R \ge D_\beta + 1,$$
$$|b_R(f) - b(f)| \le C_{\bar m,\beta}\exp[-\bar m(R - D_\beta)], \qquad R \ge D_\beta + 1.$$

The ratio N_0 is entire and even, and is strictly positive on the real axis. Its coefficients are real and obey

$$c_{2j+1} = 0, \qquad c_2 = (\log N_0)''(0), \qquad c_4 = b_{\mathrm{ch}} + 3c_2^2.$$

Uniformly in the mesh and permitted torus sizes,

$$|c_2| \le C_{\rho,\bar m,\beta} \quad (R \ge 1), \qquad |c_4| \le C_{\rho,\bar m,\beta} \quad (R \ge D_\beta + 1).$$

All error and coefficient constants are unchanged by real translations of the probe. The coefficients themselves are unchanged under physical lattice translations and cubic reflected cochains. In the charge convention $q = e\sqrt{\eta}$, $\widehat a = \eta^{1/2}a_\eta$, one has $B_t = q(t/e)\widehat a$; hence these free coefficients are independent of $e > 0$.

Proof structure and imports. The continuum calculation T02 and dilation B7 choose the positive probe. They are independent of the chamber comparison. The symbol and contact estimates S feed both the fourth-trace comparison T03 and the polarization theorem P2. The latter, together with determinant symmetry, gives T04. Only the stated free light–heavy estimates of FBM1 are imported; no running interaction, tuned bare sequence or terminal measure is assumed. In particular c_4 itself is not asserted to converge to $b(f)$: the convergent quantity is $c_4 - 3c_2^2 = b_{\mathrm{ch}}$.

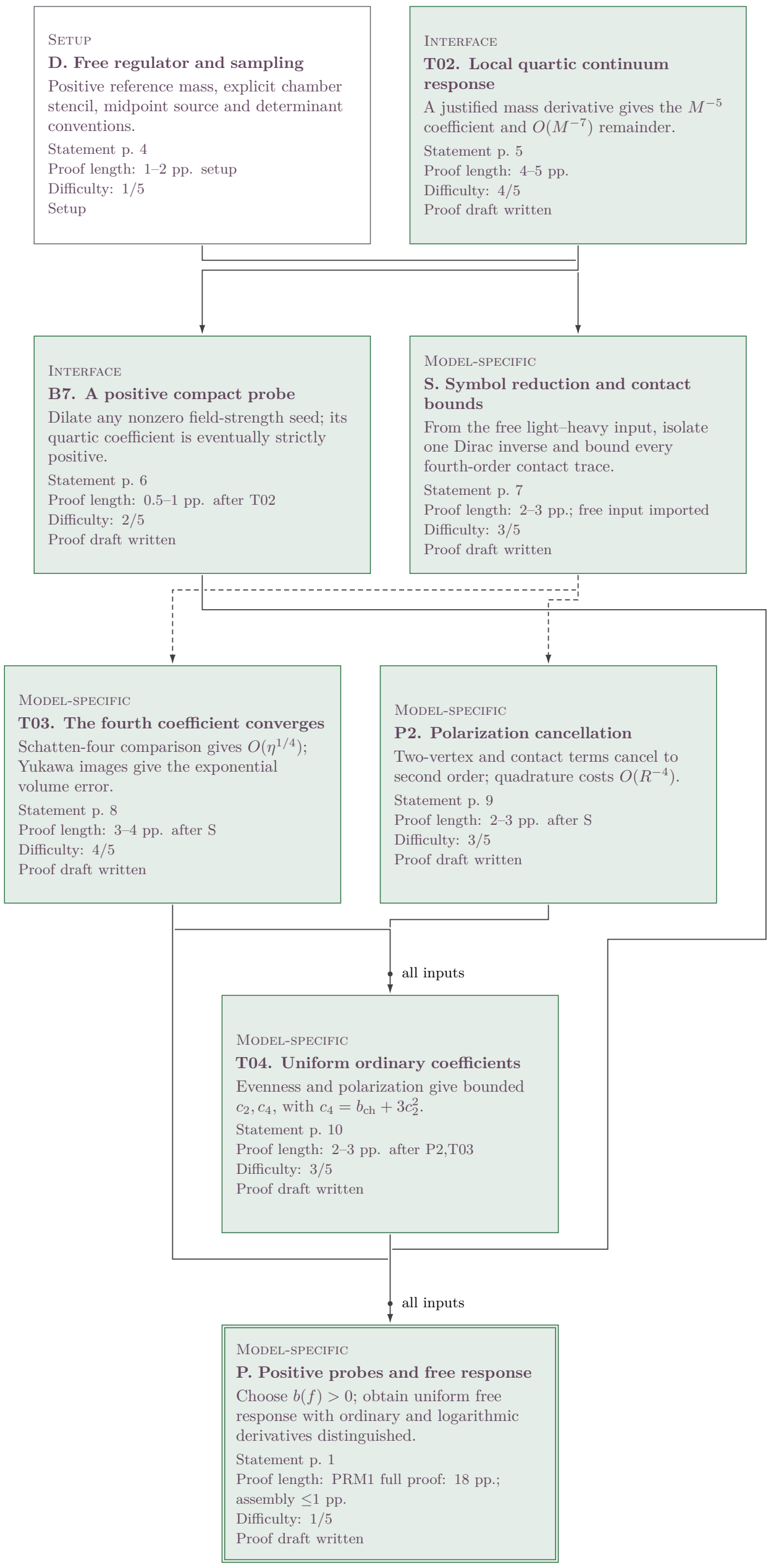
Length, difficulty and status. Full PRM1: 18 pages; difficulty 4/5. This final combination costs at most one page once its incoming results are given; its own assembly difficulty is 1/5. Proof drafted in the linked source.

Source. PRM1, Theorem A and Theorem B, pp. 1–3; their proofs occupy Sections 2–5, pp. 3–17. Broader contracts: QEO1/B7, T03 and T04.

PRO1 – Positive compact probes and free response coefficients

Proof draft written | Partial coverage | Not drafted

The continuum positivity and lattice comparison are independent proof branches. D is the one shared regulator/sampling definition. S imports only FBM1 free Schur estimates. Green means proof drafted; category text still distinguishes Setup, Interface and Model-specific statements.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.
Click a box for its statement. Return links are navigation, not proof dependencies.

T02 supplies the large-mass theorem and B7 its positive-probe consequence. B7 is not an input to convergence T03. T04 needs the independent quadratic cancellation and the fourth-trace bound. The same c4 is an ordinary derivative: only $c4-3c2^2$ converges to b(f). Difficulty rates the box's own proof given incoming results; page budgets overlap and must not be added.

2. D. The free regulator and the sampled source

Setup. Overview Target.

Continuum data. Use $\gamma_\nu = \mathrm{diag}(\sigma_\nu, -\sigma_\nu)$, $\gamma_5 = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$, and the Euclidean codifferential $(d^*\beta)_\nu = -\sum_{\mu\neq\nu} \partial_\mu \beta_{\mu\nu}$. The Fourier convention is $\widehat{v}(P) = \int v(x) e^{-iP\cdot x}\,\mathrm{d}x$. For compact real smooth β, set

$$J_\eta v(y) = \eta^{-3} \int_{[-\eta/2,\eta/2]^3} v(y+z)\,\mathrm{d}z, \qquad \delta_{\eta,\mu} v(y) = \frac{v(y+\eta e_\mu/2) - v(y-\eta e_\mu/2)}{\eta},$$
$$a^0_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu} J_\eta \beta_{\mu\nu}, \qquad a_{\eta,\nu}(\eta c) = \sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2 + Rz).$$

The unperiodized interpolant is formed before midpoint sampling. The lattice phase is $B_t(c, c+e_\nu) = t\eta a_{\eta,\nu}(\eta c)$, with the reversed-bond sign. The positive free mass is $m = \bar{m}\eta$, $0 < \bar{m} \le 1$.

Chamber operator. Let O_{cub} be the 48 signed permutation matrices. Let s_0 change the sign of coordinate 3, and s_1, s_2 swap coordinates $(2,3)$, $(1,2)$, respectively. Let r_{e_1} change the sign of coordinate 1. Define

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1,$$
$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6), \qquad M_i(g) = \tfrac{1}{2} w_i (\rho + \gamma \cdot g n_i), \quad 0 < \rho < 1.$$

For $\psi(c + ne_\nu, g) = -\psi(c,g)$, $n = R/\eta \ge 2$, put

$$(D^{\mathrm{ch}}[B]\psi)(c,g) = \sum_{i=0}^{2} M_i(g)[\psi(c,g) - \psi(c, gs_i)]$$
$$+ M_3(g)[\psi(c,g) - e^{iB(c, c+ge_1)}\psi(c+ge_1, gr_{e_1})].$$

The cochain is defined on covering oriented nearest-neighbor bonds and is $n\mathbb{Z}^3$-periodic. When $n = 2$, bonds with the same quotient endpoints remain distinct. The pairing is counting measure in c and $48^{-1}\sum_g$ in the labels. Determinants and traces are those of linear maps on the full $192n^3$-dimensional space; no factor $1/48$ multiplies a matrix trace. For the sampled phase $B_t = t\eta a_\eta$, define

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + m)}{\det(D^{\mathrm{ch}}[0] + m)}, \qquad m = \bar{m}\eta > 0.$$

Momentum conventions. Use representatives $\mathcal{P}_n = (2\pi/n)(\mathbb{Z} + \frac{1}{2})^3 \cap (-\pi, \pi]^3$ for spinors. Continuum antiperiodic momenta are $\mathcal{P}_R = (2\pi/R)(\mathbb{Z} + \frac{1}{2})^3$. The bosonic momenta in a sampled source are

$$\mathcal{K}_{\eta,R} = \{(2\pi/R)j : j \in \mathbb{Z}^3, -n/2 < j_\nu \le n/2\}.$$

Length, difficulty and status. Approximately 1–2 pages of shared setup, difficulty 1/5. Definitions; there is no separate proof claim.

Source. PRM1, Section 1.1, pp. 2–3, (3)– (6); the explicit stencil (5). This is the one shared model-definition input to the other cards.

3. T02. The local quartic continuum response

Interface. Overview Target.

Objects and hypotheses. Let $\gamma_\nu = \operatorname{diag}(\sigma_\nu, -\sigma_\nu)$, $S_M = (M + \gamma \cdot \partial)^{-1}$ on $L^2(\mathbb{R}^3; \mathbb{C}^4)$, and let $a \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ be real. Define

$$V_a = i\gamma \cdot a, \qquad q_{4,M}(a) = -\tfrac{1}{4} \operatorname{Tr}(S_M V_a)^4.$$

The available kernel is $S_M(z) = (M - \gamma \cdot \partial) e^{-M|z|}/(4\pi|z|)$. The statement concerns the doubled three-dimensional Dirac operator and an Abelian real vector potential; these conditions are retained.
Claim. For every $M > 0$, the fourth power is trace class and its trace is real. For a finite constant depending on the compact smooth potential,

$$\left| q_{4,M}(a) - \frac{1}{240\pi M^5} \int_{\mathbb{R}^3} |\operatorname{curl} a|^4 \, \mathrm{d}x \right| \le C_a M^{-7}, \qquad M \ge 1.$$

Let $R_a^{(4)}(x, y; M)$ be the kernel of $(S_M V_a)^4 S_M$. It has a continuous extension to the diagonal. Define $r_4(x; M, a) = \operatorname{tr} R_a^{(4)}(x, x; M)$, with the trace over the four spin components. Its integral is the mass derivative:

$$\frac{\mathrm{d}}{\mathrm{d}M} q_{4,M}(a) = \int r_4(x; M, a) \, \mathrm{d}x, \qquad r_4(x; M, a) = -\frac{|\operatorname{curl} a(x)|^4}{48\pi M^6} + O_a(M^{-8}).$$

The remainder is uniform on a fixed unit neighborhood of the support; the integral outside that neighborhood is $O_a(e^{-M})$.
Proof work. Use the Yukawa bound to place $S_M V_a S_M$ in the Hilbert–Schmidt class. Justify the trace and its derivative by absolutely convergent four- and five-propagator integrals. Five convolutions of the $|z|^{-2} e^{-M|z|/2}$ majorant give the continuity needed on the diagonal, also for the auxiliary linear connection. After scaling at a point, Taylor-expand through total order five. A compact gauge transformation removes the constant term and the symmetric first derivative; quartic homogeneity removes all orders below four, and parity removes order five. The surviving coefficient is evaluated with the constant magnetic-field heat kernel. Identify its formal inverse coefficients on Schwartz spinors and their continuous kernels before evaluating the diagonal. Integrate the resulting mass derivative from M to infinity. The numerical coefficient and the M^{-7} remainder are conclusions to prove, not assumptions.
Inputs. The displayed kernel, Clifford identities, elementary convolution bounds and Schatten trace operations. No lattice estimate or renormalization input.
Length, difficulty and status. Approximately 4–5 proof pages, excluding the dilation consequence; difficulty 4/5. Proof drafted.

Source. PRM1, Theorem A, Section 2, pp. 3–7. The central equations are (15), (20) and (24). This retains the large-mass part of QEO1/T02.

4. B7. Choose a positive compact probe by dilation

Interface. Overview Target.

Objects and hypothesis. Fix any $\bar{m} > 0$, the doubled Clifford matrices $\gamma_\nu = \mathrm{diag}(\sigma_\nu, -\sigma_\nu)$, and a real $\beta_0 \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ such that $f_0 = dd^*\beta_0 \not\equiv 0$. Set $a = d^*\beta_0$ and

$$I = \int_{\mathbb{R}^3} |\operatorname{curl} a|^4 \, \mathrm{d}x > 0, \quad \beta_\ell(x) = \ell^2 \beta_0(x/\ell), \quad a_\ell(x) = \ell a(x/\ell), \quad f_\ell = dd^*\beta_\ell.$$

Use the massive free kernel and the asymptotic theorem of T02; its actual input is the displayed M^{-5} term with $O_a(M^{-7})$ remainder.

Claim. The unitary dilation gives the exact identity

$$q_{4,\bar{m}}(a_\ell) = \ell^8 q_{4,\bar{m}\ell}(a).$$

For $\ell \geq 1$, $\bar{m}\ell \geq 1$, this yields

$$b(f_\ell) := -6 \operatorname{Tr}(S_{\bar{m}} i\gamma \cdot a_\ell)^4 = \frac{\ell^3 I}{10\pi \bar{m}^5} + O_a(\ell \bar{m}^{-7}).$$

Consequently every such seed gives $b(f_\ell) > 0$ for all sufficiently large ℓ. In particular a positive compact probe exists for each positive mass. The seed is nonempty: take a real bump $\varphi(x) = e^{-1/(1-|x|^2)}$ on $|x| < 1$, zero outside, and $(\beta_0)_{12} = \varphi$, with its other independent components zero. Its $\operatorname{curl} d^*\beta_0$ is nonzero.

Proof work and use. Check the dilation on the inverse and multiplication separately. The leading term grows as ℓ^3 while the error grows as ℓ. Choose this one probe before any interacting comparison. The lattice convergence in T03 applies to all compact real smooth probes; it does not require this sign theorem.

Length, difficulty and status. Approximately half to one proof page, difficulty 2/5 given T02. Proof drafted. This is a principal existence consequence, not an additional lengthy estimate.

Source. PRM1, Theorem A, Section 2.4, p.7, especially (25). The node retains QEO1/B7 and the existence conclusion of QEO1/T02.

5. S. Isolate the Dirac singularity and bound the contact terms

Model-specific. Overview Target.

Objects and precise free input. Use the chamber stencil in D. Its free symbol on the 192-dimensional fiber is $\mathcal{D}_m(p)$, with $0 < m \le 1$, $p \in [-\pi,\pi]^3$. Let $J : \mathbb{C}^4 \to \ell^2(O_{\mathrm{cub}};\mathbb{C}^4)$ send a spinor to a constant label field, and put $Q = I - JJ^*$. In the light–heavy blocks, import the following verified free estimates from FBM1:

$$H_m = D_{\mathrm{hh}}^{-1}, \quad \|H_m(p)\| \le (g_{\mathrm{h}} + m)^{-1}, \qquad \|D_{\mathrm{lh}}(p)\| + \|D_{\mathrm{hl}}(p)\| \le C_\rho |p|,$$
$$B_m = D_{\mathrm{ll}} - D_{\mathrm{lh}} H_m D_{\mathrm{hl}} = m + i\gamma\cdot p + O_\rho(|p|^2), \qquad |B_m(p)v| \ge c_\rho (m^2 + |p|^2)^{1/2}|v|.$$

All constants depend only on ρ, not on m. Only these free clauses of FBM1 Proposition 2.2 are imported. No complex-background walk or interacting activity is needed for this card.

Symbol and vertex outputs. With $R_m = \mathcal{D}_m^{-1}$, prove

$$R_m(p) = J(m + i\gamma\cdot p)^{-1}J^* + \mathcal{E}_m(p), \qquad \|\mathcal{E}_m(p)\| \le C_\rho,$$
$$\|\partial^\alpha R_m(p)\| \le C_{\alpha,\rho}(\mathrm{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|} \quad (|\alpha| \le 4).$$

The hop matrices and vertices are

$$(T_\nu^\pm u)(g) = 1_{\{ge_1 = \pm e_\nu\}} 3(\rho \mp \gamma_\nu) u(gr_{e_1}), \qquad \mathcal{D}_m(p) = \mathcal{D}_{m,\mathrm{loc}} - \sum_\nu (e^{ip_\nu} T_\nu^+ + e^{-ip_\nu} T_\nu^-),$$

$$V_\nu = \partial_\nu \mathcal{D}_m, \quad W_\nu = \partial_\nu^2 \mathcal{D}_m, \quad J^* V_\nu(0) J = i\gamma_\nu, \quad \|\partial^\alpha \mathcal{D}_m\| \le 6 \quad (|\alpha| \ge 1).$$

Mixed second symbol derivatives vanish. The involution $(\mathcal{U}u)(g) = \gamma_5 u(-g)$ obeys $\mathcal{U}\mathcal{D}_m(p)\mathcal{U} = \mathcal{D}_m(-p)$ and $\mathcal{U}V_\nu(p)\mathcal{U} = -V_\nu(-p)$.

Fourth derivative and contact bound. Use the sampled probe in D, with $A_\nu(c) = a_{\eta,\nu}(\eta c)$, $b_\nu^+ = \eta A_\nu$, $b_\nu^- = -\eta A_\nu(\cdot - e_\nu)$. For shifts $U_\nu^\pm \psi(c) = \psi(c \pm e_\nu)$, put

$$D_j = -\sum_{\nu,\pm} (i b_\nu^\pm)^j T_\nu^\pm U_\nu^\pm, \qquad G = (D^{\mathrm{ch}}[0] + m)^{-1}, \qquad Y_j = G D_j.$$

These are derivatives: the expansion has $t^j D_j / j!$. The exact logarithmic fourth derivative is

$$b_{\mathrm{ch}} = \mathrm{Tr}\, Y_4 - 4\,\mathrm{Tr}(Y_1 Y_3) - 3\,\mathrm{Tr}\, Y_2^2 + 12\,\mathrm{Tr}(Y_1^2 Y_2) - 6\,\mathrm{Tr}\, Y_1^4.$$

For $m = \bar{m}\eta$, $R \ge D_\beta + 1$, all four contact terms on the right have total absolute value at most

$$C_{\rho,\beta}\eta[1 + \log(1 + 5/(\bar{m}\eta))].$$

The constant is independent of $\eta, R, \bar{m}$ and the position of the probe.

Proof work. Apply block inversion. Each mixed block supplies $|p|$, which cancels the light singularity; the quadratic Schur remainder leaves a bounded inverse difference. Differentiate the inverse identity for the derivative estimates. Derive all phase vertices with their factorial conventions. Cube-average sampling and Poisson summation give uniform ℓ^1 and supremum Fourier bounds for powers of A. In each cyclic contact trace keep one Fourier factor in supremum norm and the others in ℓ^1; the remaining momentum sum has at most three inverse factors. Its worst radial integral is logarithmic, producing the displayed bound.

Length, difficulty and status. Approximately 2–3 proof pages, excluding the imported FBM1 proof; difficulty 3/5. Proof drafted.

Source. PRM1, Section 3, pp. 8–10; (26)– (37). The verified import is FBM1 Proposition 2.2.

6. T03. The chamber fourth coefficient converges to the continuum trace

Model-specific. Overview Target.

Objects and hypotheses. Fix $0 < \rho < 1$, $0 < \bar{m} \le 1$, and a real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$. Let $a = d^*\beta$, $f = dd^*\beta$, and let D_β be the least side of a support cube. Use the explicit phase sampling and chamber operator in D, $m = \bar{m}\eta > 0$, $0 < \eta \le 1$, $n = R/\eta \in \mathbb{N}$, $n \ge 2$, $R \ge D_\beta + 1$. Assume the bounded light-inverse remainder, normalized first vertex, inverse bounds and fourth-contact bound displayed in S. These are the exact symbol and finite-trace estimates used here; no sign of $b(f)$ is assumed. Define

$$b_{\mathrm{ch}} = \partial_t^4 \log \frac{\det(D^{\mathrm{ch}}[t\eta a_\eta] + m)}{\det(D^{\mathrm{ch}}[0] + m)}\bigg|_{t=0},$$

$$b_R(f) = -6\,\mathrm{Tr}(S^R i\gamma \cdot a^R)^4, \qquad b(f) = -6\,\mathrm{Tr}(S_{\bar{m}} i\gamma \cdot a)^4.$$

Here a^R is the periodization of a, and S^R is the massive Dirac inverse on antiperiodic momenta $(2\pi/R)(\mathbb{Z} + \frac{1}{2})^3$.

Claim. Both continuum operators belong to the Schatten class of order four, their fourth traces are real, and

$$|b_{\mathrm{ch}}(\eta, R) - b_R(f)| \le C_{\rho,\bar{m},\beta}\eta^{1/4}, \qquad |b_R(f) - b(f)| \le C_{\bar{m},\beta} e^{-\bar{m}(R - D_\beta)}.$$

Constants are unchanged by real translations of β. Consequently $b_{\mathrm{ch}} \to b(f)$ whenever $\eta \to 0$, $R \to \infty$. This is a logarithmic fourth coefficient statement, not convergence of $N_0^{(4)}(0)$.

Proof work. For a Fourier multiplier and a scalar multiplication operator prove

$$\|M_h v\|_4 \le \left(n^{-3} \sum_{p \in \mathcal{P}_n} \|h(p)\|_4^4 \right)^{1/4} \|v\|_{\ell^4},$$

and its continuum torus version. Use the nonnegative trace of a commutator square, then truncate momenta for the infinite-rank version. The whole-space Yukawa calculation gives $\|S_{\bar{m}} i\gamma \cdot a\|_4^4 \le (2\pi\bar{m})^{-1} \int |a|^4$. Use Schatten Hölder and the telescoping fourth-power estimate

$$|\,\mathrm{Tr}\, X^4 - \mathrm{Tr}\, Z^4| \le 4 \max(\|X\|_4, \|Z\|_4)^3 \|X - Z\|_4.$$

After the contact bound, make five comparisons: remove the bounded inverse part; replace the compressed hop by multiplication; compare sampled and interpolated Fourier coefficients; replace the interpolant by a^R; remove the high-momentum cutoff. Their errors are respectively $O(\eta^{1/4})$, $O(\eta^{1/4})$, $O(\eta^2)$, $O(\eta)$, $O(\eta^{1/4})$, uniformly in the torus side. Finally compress both continuum traces to the support cube and sum the nonzero antiperiodic Yukawa images. Its Hilbert–Schmidt kernel difference is exponentially small in $R - D_\beta$.

Inputs. The concrete finite free estimates in S, the sampling of D, the massive Dirac kernel and elementary Schatten identities. The positivity theorem B7 is independent of this proof.

Length, difficulty and status. Approximately 3–4 proof pages, excluding S; difficulty 4/5. Proof drafted.

Source. PRM1, Theorem B, Section 4, pp. 10–13, especially (8) and (38)– (45).

7. P2. The polarization cancellation and finite-volume quadrature

Model-specific. Overview Target.

Objects and explicit hypotheses. Use the chamber symbol $\mathcal{D}_m(p)$ of D, $R_m = \mathcal{D}_m^{-1}$, $V_\nu = \partial_\nu \mathcal{D}_m$, $W_\nu = \partial_\nu^2 \mathcal{D}_m$, with mixed second derivatives zero. Fix $0 < \rho < 1$, $0 < \bar{m} \le 1$, $m = \bar{m}\eta > 0$, $0 < \eta \le 1$, $n = R/\eta \ge 2$ integral. From S, use

$$\|\partial^\alpha R_m(p)\| \le C_{\alpha,\rho}(\operatorname{dist}(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|} \quad (|\alpha| \le 4),$$

bounded symbol derivatives, and the involution identities $\mathcal{U}\mathcal{D}_m(p)\mathcal{U} = \mathcal{D}_m(-p), \mathcal{U}V_\nu(p)\mathcal{U} = -V_\nu(-p)$. For $\mu, \nu \in \{1, 2, 3\}$, set

$$\begin{aligned} L_{\mu\nu}(p,k) &= \delta_{\mu\nu} \operatorname{tr}[R_m(p)W_\nu(p)] \\ &\quad - \operatorname{tr}[R_m(p)V_\mu(p + k/2)R_m(p + k)V_\nu(p + k/2)], \end{aligned}$$

$$\Pi_{\mu\nu}^{\eta,R}(K) = \eta^{-1}n^{-3} \sum_{p \in \mathcal{P}_n} L_{\mu\nu}(p, \eta K), \qquad \Pi_{\mu\nu}^{\eta}(K) = \eta^{-1} \int_{[-\pi,\pi]^3} L_{\mu\nu}(p, \eta K) \frac{\mathrm{d}p}{(2\pi)^3}.$$

Here $\mathcal{P}_n = (2\pi/n)(\mathbb{Z} + \frac{1}{2})^3$ modulo 2π. The first function is a finite-volume polarization; the second retains the lattice symbol but has no volume parameter.

Claim. For all real K,

$$|\Pi_{\mu\nu}^{\eta,R}(K) - \Pi_{\mu\nu}^{\eta}(K)| \le C_\rho \bar{m}^{-3} R^{-4},$$
$$|\Pi_{\mu\nu}^{\eta}(K)| \le C_\rho \bar{m}^{-1}|K|^2, \qquad |\nabla_K \Pi_{\mu\nu}^{\eta}(K)| \le C_\rho \bar{m}^{-1}|K|.$$

The constant depends only on ρ. For the midpoint source transform

$$h_{\eta,\nu}(K) = \eta^3 \sum_{y \in \eta\mathbb{Z}^3 + \eta e_\nu/2} a_{\eta,\nu}^0(y) e^{-iK\cdot y},$$

the exact second logarithmic coefficient is

$$(\log N_0)''(0) = R^{-3} \sum_{K \in \mathcal{K}_{\eta,R}} \sum_{\mu,\nu} \overline{h_{\eta,\mu}(K)} h_{\eta,\nu}(K) \Pi_{\mu\nu}^{\eta,R}(K).$$

This identity needs no separation from periodic images of the probe.

Proof work. Keep the two-vertex and contact traces together, derive the midpoint phase and the exact scalar factor $R^{-3}\eta^{-1}n^{-3}$. Four integrations by parts in a largest Fourier coordinate give an antiperiodic quadrature error bounded by $42n^{-4}$ times the integrated fourth derivative. The inverse estimates bound that integral by $C_\rho m^{-3}$; use $\eta^{-1}n^{-4}m^{-3} = \bar{m}^{-3}R^{-4}$. At zero external momentum the whole integrand is $\partial_\mu \operatorname{tr}(R_m V_\nu)$, whose period integral vanishes. Symmetric midpoint vertices and parity remove the linear term. The integrated Hessian is bounded by $C_\rho m^{-1}$. Taylor's formula supplies both external momentum powers, and $\eta/m = \bar{m}^{-1}$ gives the claimed physical bound.

Length, difficulty and status. Approximately 2–3 proof pages, difficulty 3/5. Proof drafted. The statement remains the concrete chamber verification; no new general-symbol theorem is claimed.

Source. PRM1, Proposition 5.1, Section 5.2, pp. 14–16, (51)– (61).

8. T04. Symmetry and bounded ordinary determinant coefficients

Model-specific. Overview Target.

Objects and hypotheses. Let D^{ch}, $a^0_{\eta,\nu}$, its midpoint periodization a_η, and $N_0(t)$ be the explicit finite objects in D. Fix $0<\rho<1$, $0<\bar m\le 1$, $0<\eta\le 1$, $n=R/\eta\in\mathbb{N}$, $n\ge 2$, and a real nonzero compact smooth β. The free mass $m=\bar m\eta$ is positive. Set

$$c_j=N_0^{(j)}(0),\quad a_{\mathrm{ch}}=(\log N_0)''(0),\quad b_{\mathrm{ch}}=(\log N_0)^{(4)}(0),$$

using the logarithm equal to zero at the origin. Use the exact second-coefficient representation and the three polarization bounds in P2. For the fourth coefficient, also use T03's finite continuum trace and its two explicit error bounds, on $R\ge D_\beta+1$.

Claim. The determinant ratio is entire and even, strictly positive for real t, and

$$c_j\in\mathbb{R},\qquad c_{2j+1}=0,\qquad c_2=a_{\mathrm{ch}},\qquad c_4=b_{\mathrm{ch}}+3c_2^2.$$

There is a constant depending only on $\rho,\bar m,\beta$, unchanged by all real translations of the probe, such that

$$|c_2|\le C_{\rho,\bar m,\beta}\quad(R\ge 1),\qquad |c_4|\le C_{\rho,\bar m,\beta}\quad(R\ge D_\beta+1).$$

The coefficients are independent of physical charge e when $B_t=q(t/e)\widehat a$, $q=e\sqrt{\eta}$, $\widehat a=\eta^{1/2}a_\eta$. They are exactly unchanged under lattice translations and cubic reflections of the external cochain. Uniformity of constants under all real probe translations is a separate assertion, not exact translation invariance of the sampled coefficient.

Proof work. The doubled spin sectors are adjoint at real phase, and $\mathrm{Re}(D^{\mathrm{ch}}[B]+m)\ge mI$; hence the real determinant is a strictly positive squared modulus. The charge matrix $C=\sigma_2\oplus\sigma_2$ gives $C(D^{\mathrm{ch}}[B]+m)^{\top}C^{-1}=D^{\mathrm{ch}}[-B]+m$, also for complex phases. Finite phase exponentials give entirety and charge conjugation gives evenness. Expanding the local logarithm then gives the displayed ordinary/logarithmic coefficient identities. The spin lift of a cubic reflection gives similarity of the phased operators and preserves the antiperiodic spin space. For the bounds, discrete integration by parts proves, for every integer $j\ge 0$,

$$|h_{\eta,\nu}(K)|\le C_{\beta,j}(1+|K|)^{-j},\qquad |K|_\infty\le\pi/\eta,$$

with the explicit permissible choice $C_{\beta,j}=2\cdot 12^j(D_\beta+j+3)^3|\beta|_{C^{j+1}}$. Pair this with P2: use $j=2$ for the quadrature term and $j=3$ for its quadratic momentum term. The sum $R^{-3}\sum_K(1+|K|)^{-4}$ is bounded uniformly for $R\ge 1$. This proves the second-order bound. The fourth-trace comparison and $c_4=b_{\mathrm{ch}}+3c_2^2$ give the fourth-order bound. Estimating the original two traces separately does not prove the uniform second-order statement.

Length, difficulty and status. Approximately 2–3 proof pages including symmetry and discrete probe decay, excluding P2 and T03; difficulty 3/5. Proof drafted.

Source. PRM1, Section 5.1, pp. 13–14 and Section 5.3, p.17; Proposition 5.2, with (46) and (62). This is the completion of Theorem B.